A Survey of Melody Extraction Techniques for Music Information Retrieval

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Background in musicology. Most of the time the concept of melody is associated to a monophonic sequence of pitch notes. Also, identifying the main melody line in music, monophonic sequence of notes can be further deconstructed using the melody feature selection techniques in music information retrieval.

Background in computer science. Some computational research in musical structure are concerned with building information retrieval systems, defining similarity measures, and otherwise finding occurrences and variations of a musical fragment within a collection of music documents. A very few of them includes extraction of melody, which is to extract and represent the melodic content of the music data. The purpose of a melody extraction technique is to identify sequences of notes that are likely to correspond to the perceived melody, given the volumes of music available online. It is not practicable to do this by hand. In terms of the needs of the user, a user may wish to query on a distinctive musical pattern that occurs in some parts.

Aims. This survey provides an overview of melody extraction techniques for music information retrieval for audio (waveform representation) and symbolic music notation (symbolic representation). Methods and extraction techniques are briefly presented. We have placed techniques on a test-bed which is created some melodic structure in music and we research that the simplest symbolic representation techniques to cover between the melodic movements in musical structure. As a result the simplest of these, Skyline and Best Channel starting at any instant time are the best method to the other more sophisticated one.

Most of the time the concept of melody is associated to a sequence of pitch notes. This definition can be found: "A combination of a pitch series and a rhythm having a clearly defined shape" (Solomon 1996), and on Grove Music: "Pitched sounds arranged in musical time in accordance with given cultural conventions and constraints" (Grove's online 2002). In the definition proposed in (EDT 2002), melody is defined with some of these different connotations: As a set of musical sounds in a pleasant order and arrangement as a theme in a music structure. As shown in EDT definition, melody can be also related to the concept of tune or predominant theme that is usually defined as something recurrent, neither too short nor too long, where all the notes are usually played by the same instrument nor it is often located at the top of the register.

Recently, some computational research connecting with musical structure such as Music Information Retrieval (MIR) are concerned with melody, defining similarity measures between melodies, and otherwise finding occurrences and variations of a musical structure within a melody. A very few of them includes extraction of melody, which is to extract and represent the melodic content of the music data.

Because of the fact that polyphonic nature of music structure permits simultaneous sound of notes, using the polyphonic music in MIR is able to computationally expensive. On the other hands, in terms of the needs of the user, a user may wish to query, in MIR systems, as monophonic instead of polyphonic.

The majority of the methods suggested fall into two frameworks in order to use music in

MIR. The first framework models music as linear strings. The second one models music as sets of two-dimensional geometric objects. Although Lemström's study explains that the geometric framework is the choice for the music information retrieval task (Lemström 2007), a lot of studies about reduction to monophonic from polyphonic use music as a linear string because of the fact that monophonic music consists of linear note sequence and it is more appropriate for music retrieval.

Melody extraction is a process, which generates monophonic equivalent of polyphonic files. Hence, output of melody extraction takes less space in databases by containing only genuine part of the music. Moreover, retrieval on generated monophonic files can be executed in a reasonable time.

After definition of melody and melody extraction shortly, in this paper, we investigate some of working which have been done in the past deal with researching of melody extraction both from a collection of music recorded as MIDI and audio files. Especially, in particular, we try to explain how implemented the methods of MIDI file melody extraction. Our work makes the following sections: Firstly, related works on melody extraction that all of the searching deals with melody extraction both waveform and symbolic representation. Secondly, evaluation and experimentation of the melody extraction methods are presented and test results are discussed. Finally, conclusion of the paper gives a look to the further studies on this subject.

Related Works on Melody Extraction

Up till now, there have been two general basic rules for melody extraction. Firstly, listeners usually hear the highest pitch notes as melody (Franoes 1958), except when they are monotonous. So, a melody is heard as the figure if it is higher than the accompanying parts. Secondly, pitch proximity is the most important factor in grouping notes into perceived parts (Deutsch 1999). Consequently, is it possible to extract melodies from existing digital music corpora by machine and use them to build a melody database for the retrieval purpose? There could be two kinds of sources from which we extract melodies and build using melody database: Waveform representation (e.g., audio and acoustical data) or symbolic representation (e.g., scores and MIDI).

Waveform Representation

Compared with symbolic; waveform representation applications are more complex to implement (Gutiérrez 2002). Also, before melody extraction in audio files, certain steps should be passed. These steps can be enumerated on:

1. Pitch Estimating
2. Note Segmentation
3. Extracting Melody from Segmented Note Sequences

But some of the related works on audio extraction of the melody can be continued:

4. Melodic Segmentation
5. Melody Pattern Extraction
6. Melodic Transformation

Pitch estimating, called finding fundamental frequency, is the main descriptor to be considered when describing audio melody extraction. The first solution adopted for fundamental frequency detection of musical signals was to adapt the techniques proposed for speech (McLane 1996). Later, other methods have been specifically designed for dealing with music (Hess 1983), (Maher 1993), (Anderson 1997), (Klapuri 2002).

Note segmentation, once the fundamental frequency has been estimated, needs to delimitate the note boundaries, in order to extract a sequence of notes or melody and the descriptors associated to note segments (McNab 1996), (Rossignol 2002), (Klapuri 1999), (Herrera 2001).

Extracting melody from segmented note sequences, considering a segmented note as a pitch sequence, is applied to complete melody extraction in audio files. As output, the methods provided monophonic pitch sequences called melody. At this point, the most important researches of melody extraction in audio files as a result of audio melody extraction are (Nettheim 1992).

Melodic segmentation, as those defined by Lerdahl and Jackendoff (1983), is to establish a structure on a sequence of notes. This structure is specified by a set of pair of boundaries that divides the audio excerpt in a series of audio segments. It may involve different levels of hierarchy, and may include overlapping segments and unclassified areas that do not belong to any segment.

Research	Explanation
Marsden (1992)	He applied the proximity principle in a rule-based approach, and attempt to refine the rules to split the parts of a Bach fugue [34].
Anderson (1997)	He described how some of the methods applied to monophonic fundamental frequency
Selfridge (1998)	He developed the pitch estimeting techniques while making a melodic comparision [40].
Klapuri (2000)	He designed a new estimation algorithm using audio spectral place to sequence notes of music [44].
Durey (2001)	Applyed audio retrieval “Wordspotting” techniques to the problem of melody extraction from collections of audio files [47].
Marolt (2006)	Finding significant melodic fragments throughout the analyzed piece of music and clustering these fragments according to their timbral similarity.
Paiva (2006)	detection and determination of pitches, and identification of melodic notes in audio
MIREX (Music Information Retrieval Evaluation eXchange)	Identifying the melody pitch contour from polyphonic musical audio using voice and pitch detection

Table 1. Some important researches about melody extraction in waveform representation and their explanation.

Melody pattern extraction, comparing two melodies and extracting a musical pattern from a sequence have common aspects, is associate the concept of melody matching to the notion of pattern induction (or extraction).

Melodic transformations, known a pitch shifting, consist on augmenting or diminishing the pitch values of the melody by a given interval. In a given key, this could be done with the constraint to preserve the scale and, of course, relevant properties as timbre and rhythm.

Symbolic Representation

Symbolic representation applications are less complex than waveform in both all of music information retrieval researches and melody extractions. Because, symbolic corpus including a data. Consequently, very important values for melody extraction such as the end of the notes, start of the notes, pitch value etc. can be find in this data.

Skyline, usually it is named *all-mono*, is proposed based on simple rule of symbolic representation by Uitdenbogerd (1999), which extracts the highest pitch line as the melody line. Concisely, the basic structure of the method is: For each onset time (a point on the time axis with at least one note onset) choose the note with the most distinct feature (e.g., highest pitch). If necessary, this note’s duration is shortened such that it does not exceed the time between the actual and the next onset time. In principle, all variants of the pitch based skyline method extract sequences from the higher pitched regions of a piece of music. Frequently, the computed extract jumps between a primary melody and accompanying notes.

Figure 1- Polyphonic music notation (A). Its Skyline results (B).

Advantage of the skyline is that the majority of melody which will be selected as a line is often the highest part in pitch. In contrary, disadvantage of that is that it will rarely select the melody cleanly from piece of music. It will not select the melody at all if it is lower in pitch than other.

Uitdenbogerd has discovered three more methods except skyline and all of them seem to be different structure or including an alternative method in respect of using the skyline. These methods' names are:

- Top Channel
- Entropy Channel
- Entropy Part

Top Channel keeps a channel with highest average pitch and it is removed all other notes sound simultaneously except for the highest note. The top channel method makes use of standard structure of MIDI files (track and channels) by examining each channel separately. Each channel is processed by applying the all-mono method to it to produce a melody for the channel. The channel melody with the highest average (mean) pitch is then chosen as a melody. Each note in the channel melody is given equal weighting in the calculation, regardless of duration. Disadvantage of Top Channel is that, in practice, sometimes wrongly identify high accompanying part. If it moves from one part to another, it is also fail to select the entire melody.

Figure 2- Polyphonic music notation (A). Extracted melody using Top Channel Method (C)

Entropy-Channel keeps the highest note of each channel and it is selected the channel with largest entropy. Firstly, it applies the skyline to each individual channel of the piece, and then the channel with the highest first-order predictive entropy is selected as a melody.

Entropy-Part uses a heuristic to segment each channel into parts and selects the highest entropy parts. This method uses timing and pitch proximity information to split the music into parts. The notes from each channel is analyzed in turn and allocated an appropriate part. If the current note accused at the same time as those in the existing parts then a new part was created. Otherwise the note was allocated to the part that was most similar in pitch, using the most recent note in each part to make that decision. Disadvantage of Entropy-Part is that if there are multiple notes occurring at precisely the same time, as happens with non-natural performance information, then method may allocate notes in unexpected ways.

Advanced Skyline, or it is known as *Advanced All-Mono*, is developed by Shan (2002). This method removes channels of instruments for each measure, and selects the channel of the largest volume. The reason is that volume of melody is usually the largest, and it does not change channel constantly, so melodies of first-half and second-half measure are in the same channel. For the selected channel of each measure, it is kept the highest note if several notes sound simultaneously.

Partitioning, or *Melody-Lines*, is also proposed by MidiLib Project Team (2000), which aims at partitioning the polyphonic piece into a set of monophonic melodies. Similar to the Skyline, the partitioning method first runs through the MIDI notes in ascending time order. It is existed melody line which is "inaudible" at time stamp. In this, inaudible at a time stamp means that the last note of that melody line must have ended before that time stamp. A basic criterion for the assignment of notes to melody lines is the minimization of pitch differences of subsequent notes. Notes that cannot be assigned to existing melody lines form new melody lines. In extending existing melodies, Team only append one note to one melody at one time stamp. Furthermore one has to take care not to introduce long intervals containing pauses.

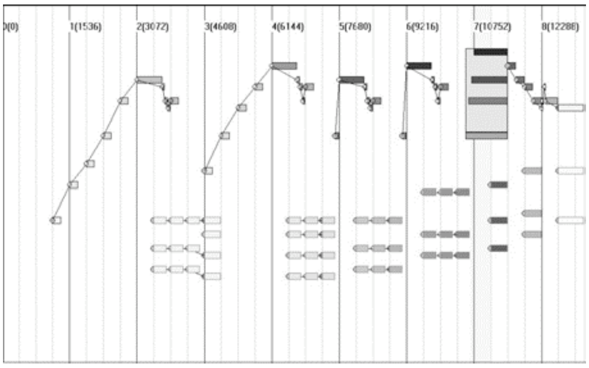

Figure 3- The partitioning method on piano-roll in MIDI

Revised Skyline developed by Wei Chai (2000) to extract melody from MIDI files with separate polyphonic melody track is a variation of the skyline method. The revised skyline method adds one parameter called time overlap parameter, which can make the recognized melody much cleaner than the classical skyline method based on the test set provided by Uitdenbogerd.

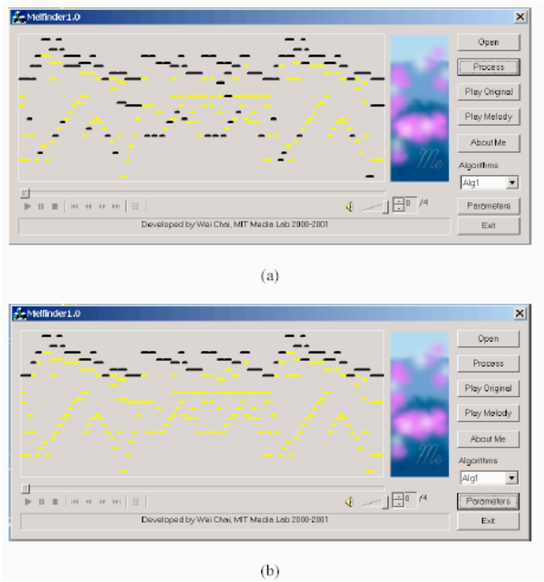

Figure 4- Results of the skyline algorithm (a) and the revised skyline algorithm (b) on piano-roll in MIDI (extracted melody contour is blank)

The Best Channel, also our approach, selects all the channels involving melody used clustering technique (Ozcan, Isikhan & Alpkocak 2005). Our approach is based on the elimination of MIDI channels those do not contain melodic information. First, MIDI channels are clustered depending on pitch histogram. Afterwards, a channel is selected from each cluster as representative and remaining channels and their notes are removed. Finally, Skyline method is applied on the modified MIDI set to ensure accuracy of monophonic melody.

Some of researches on MIR can be applied for melody extractions apart from methods that have been mentioned above. Even though these methods include melody extraction techniques, generally it is known as melody channel identification. For example, a method of splitting polyphonic music into parts uses note proximity rules (Blackburn 1998). Tang's work (2002) uses 5 different parameters like average velocity, polyphonic to monophonic time ratio, silence ratio, range name and track name. It uses attributes which are performer specific. David (2006) built a melody line identification system which uses a toolkit, and a random forest classifier for learning the track information. Colin Meek (2001) has created Melodic Motive Extractor (MME) that identifies musical keywords or themes. The system searches for all patterns composed of melodic repetition in a piece. Velusamy created a method to identify the melody track/channel in a polyphonic MIDI file (Velusamy 2007).

Evaluation and Experimentation

In order to evaluate melody extraction methods in symbolic representation, we prepared a MIDI test bed which covers fundamental music styles. Selected music styles are melody in high pitch, accompany in high pitch, melody change instrument and so on. Our MIDI test bed consists of 23 music files. All files are selected ourselves. The entire test bed can be seen in Appendix and our melody extraction application has been written as a java for visualization purposes as a supplementary tool.

We started our experiments with selecting MIDI music files. While selecting music, especially, we prefer to take musical structure which is opposite of extraction method above. By means that, if skyline selects the highest pitches as a melody, we use music with melody that is not the highest pitches, or Top Channel selects only one channel which consist of a melody, we use the music with melody that moves to a channel from another.

Because of both implementation of waveform representation is more complex and we have used already to compare some MIDI music files (Ozcan-Isikhan-Alpkocak 2005), we compare symbolic representation methods instead of waveform.

We recorded the experimental outputs of our approaches and make comparisons. Recall and precision metrics are used to evaluate melody extraction methods used in this experimentation. Meanwhile visual and auditory techniques are used to compare all Melody Extraction Algorithms. Meanwhile visual and auditory techniques are used to compare all Melody Extraction Methods.

Methods	Recall	Precision
Skyline	0.593	0.935
Best Channel	0.769	0.653
Top Channel	0.465	0.741
Entropy Channel	0.477	0.774
Entropy Part	0.377	0.385
Revised Skyline	0.234	0.606

Table 2. Performances evaluation of some melody extraction methods used in experimentation

Experiments indicate that Skyline overtakes melody extraction algorithms, when multiple channels contain melody. However, if accompany notes have high frequency, then both Skyline may eliminate melody notes. Moreover, Skyline performance decreases when data set consist of music styles which are rapid succession of notes rather than concurrent notes. In other words, in this experiment we show that Skyline is of service to use with Best Channel.

Conclusion

There didn't appear to have been much research on the extracting a melody from piece of music and several algorithms that extract melodies from polyphonic piece of music were presented. Those algorithms explained in this paper; most commonly implemented one was Uitdenberg's melody extraction algorithm. One hand, encountering some useless results and impossibility of implementing them in all cases are the main disadvantages of these algorithms, on the other hand, each one was based on aspects of simultaneous note perception. Generally, the simplest of these, that selects the highest pitch note starting at any instant worked the best compared to the other more sophisticated algorithms when evaluated by listeners. None of the algorithms can reliably extract melodies from all types of music, but this is likely to be impossible in the future. We hope that our research would contribute to widen this perspective both musicology and computer science, and for future works.

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References

Anderson, E. J. (1997). *Limitations of short-time Fourier transforms in polyphonic pitch recognition.* Technical Report Ph.D. qualifying project report, Department of Computer Science and Engineering in the Graduate School of the University of Washington

Blackburn S. & DeRoure D. (1998). *A Tool For Content Based Navigation Of Music*, Proc.of ACM Multimedia98, September 11, England

Chai, Wei. (2000). Melody *Retrieval on the Web.* MS Thesis, Massachusetts Institute of Technology, USA

David R., Pedro J., Antonio & Jose (2006). *Melody Track Identification in MIDI Files*, American Association for Artificial Intelligence

Deutch, D. (1999). *Grouping mechanisms in music.* In The Psychology of Music, chapter 4, pages 99-134

Durey, A. S. & Clements, M. A. (2000). *Melody spotting using hidden Markov models.* ISMIR 2001, p.109–117

EDT 2002. *Educational Dictionary-Thesaurus* http://www.wordsmyth.net

Franoes, R.(1958). La Perception de la Musique, Hillsdale, New Jersey, Translated by W.J. Downling (1988)

Grove's on line (2002). *Integrated music resource on the web*, http://www.grovemusic.com

Gutiérrez, E. G. (2002). *Melodic Description of Audio Signals for Music Content Processing*, research work, PhD Program

Informática i Comunicació Digital, Universitat Pompeu Fabra, Bieni

Herrera P. & Gutiérrez,E. G. (2001). Study report on audio segmentation. Technical report, CUIDADO internal report, Music Technology Group, Pompeu Fabra University

Hess, W. (1983). *Pitch Determination of Speech Signals. Algorithms and Devices.* Springer Series in Information Sciences. Springer-Verlag, Berlin

Klapuri A. (1999). Sound onset detection by applying Psychoacoustic knowledge. In IEEE International Conference on Acoustics, Speech and Signal Processing

Klapuri, A. (2000) *Qualitative and quantitative aspects in the design of periodicity estimation algorithms.* In European Signal Processing Conference

Klapuri, A. (2002). *Qualitative and quantitative aspects in the design of periodicity estimation algorithms.* In European Signal Processing Conference

Lerdahl, F., & Jackendoff, R. (1983). A generative theory of tonal music. Cambridge, Mass.: MIT Press.

Lemström K. & Pienimäki A. (2007). *On comparing edit distance and geometric frameworks in content-based retrieval of symbolically encoded polyphonic music.* Musicae Scientiae, Discussion Forum 4a: 135-152

Maher, R. C., & Beauchamp J. W. (1993) *Fundamental frequency estimation of musical signals using a two-way mismatch procedure.* Journal of the Acoustic Society of America, v.95:2254–2263

Marolt, M. (2006). *A mid-level melody-based representation for calculating audio similarity.* Proc. ISMIR 2006

Marsden, A. (1992). *Modelling the perception of musical voices: a case study in rule-based systems.* In Computer Representations and Models in Music, pages 239-263.

McLane, A. (1996). *Music as information*. Annual Review of Information Science and Technology 31, p.225–262.

McNab, R. J. & Smith, L. A. & Witten I. H. (1996) Signal processing for melody transcription. SIG, Working paper, 95(22).

Meek, C. & Birmingham, W. P. (2001). *Thematic Extractor.* Proceeding in ISMIR 2001, p.119–128.

MidiLib Project Team. *http://www-mmdb.iai.uni-bonn.de/forschungprojekte/midilib/english/skydemo.html*

Nettheim, N. (1992). On the spectral analysis of melody. Journal of New Music Research, v. 21:135–148

Ozcan G., Isikhan C. & Alpkocak A. (2005). *Melody Extraction on MIDI Music Files*, Proc.of the Seventh IEEE International Symposium on Multimedia (ISM05).

Paiva R.P., Mendes, T. &Cardoso A. (2006). *Melody Detection in Polyphonic Musical Signal: Exploiting Perceptual Rules, Note Salience, and Melodic Smoothness.* Computer Music Journal. V.30, issue 4, winter, pp.80-98

Rossignol, S. (2002). Segmentation et indexation des signaux sonores musicaux. Phd thesis, Université Paris VI- IRCAM

Selfridge Field, E. (1999). *Conceptual and Representational Issues in Melodic Comparison. Melodic Similarity: Concepts, Procedures, and Applications.* MIT Press, Cambridge, Massachusetts

Shan, M. K. & Kuo, F. - F. (2002). Music Style Mining and Classification by Melody. IEICE Trans. inf. & syst., v.85 no.1

Solomon,L.(1996).http://www.azstarnet.com/solo/glossary.htm

Tang M., Yip C. L. & Kao B. (2002). *Selection of Melody Lines for Music Databases*, In IEEE Proc. of the 24th Annual International Computer Software and Applications Conference, 2000.

Uitdenbogerd, A. & Zobel, J. (1999). *Melodic matching techniques for large music databases.* In Proceedings of ACM International Multimedia Conference, Orlando Florida, USA

Velusamy S., Thoshkahna B. & Ramakrishnan K.R. (2007). *A Novel Melody Line Identification Algorithm*, Lecture Notes in Computer Science, Springer-Verlag, Berlin

ID	Name	Melody Channel	METHODS					
			Best Channel	Skyline	Top Channel	Entropy Channel	Entropy Part	Revised Skyline
1	Mozart - Alla Turka	2	1,2	1	1	1	1	1
2	Chopin - Etud Op 12	1	1,2	1	1	1	1	1
3	Mozart -12 V. (theme)	2	1,2	1	1	2	1	1
4	Beethoven 5. Symph	12	4,12,3	1	1	11	2	2
5	Mozart -12 V. (6V)	2	1,2	1	1	2	2	1
6	Mozart-Concerto	9	1,3,2	1	2	1	6	3
7	Madonna - Frozen	11	4,12,2	12	11	4	4	11
8	R.Martin - Maria	8	6,7,3	6	9	6	6	6
9	Bach - Goldberg	1	1	1	1	1	1	1
10	Barry M. - C.Cab.	3	2,7,3	7	7	2	2	7
11	Yanni - Themes	4	12,1,2	1	1	1	1	1
12	Green Sleeves	1	1	1	1	1	1	1
13	Rodgers - Romantic	3	4,1	1	1	1	1	1
14	Albeniz - Asturias	2	1,2	2	1	2	2	1
15	Pulp Fiction	4	5,6,3	6	6	5	3	6
16	Empire Strikes Back	1	3,4,1	3	9	3	8	3
17	Tschaikovski - 1812	15	1,7,15	1	1	1	11	1
18	F. Lizst - Totantanze	4	2,3,1	4	1	4	3	4
19	M. Jackson - Thriller	7	4,6,3	4	7	4	2	2
20	C. Dion - All By M.	5	1,7,8	7	7	1	2	7
21	Eric Clapton-Tears in h.	5	5,7,2	5	5	3	5	3
22	Celine Dion- All By M.	1	6,1,8	1	15	1	13	1
23	Bon Jovi - Always	1	1,6,3	1	8	1	11	1

Appendix: Test-bed used for evaluation. Melody Channel Column consist of a melody of test song that is given ID and Name in first and second column. Remaining columns called methods denote the selected channels of specified melody extraction methods.